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Designing PIDs for Reproducible Science Using Time-Series Data

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Abstract. The use of persistent identifiers (PIDs) is seen as a part of the solution to the problem of reproducible research and science. This short paper proposes a preliminary method using PIDs to reproduce research results using time-series data. Furthermore, we feel it is possible to use the methodology and design for other types of datasets.

Keywords: PID · ARKs · time-series · datasets · reproducibility

1 Introduction

Reproducible research and science has become an urgent problem as machine learning, and Ai algorithms become more complex (e.g., deep learning). Part of this reproducibility problem is the lack of specificity as to the exact data used for training and testing these learning systems. Part of the investigative work done by IEEE Standards Association P2957 Data Governance and Metadata Management Working Group (BDGMMWG) is to look at how to create a persistent identifier (PID) system that point to specific data from a given data repository.

2 Methodology

A federated metadata registry (as in Fig. 1) is an extensive database that stores and manages datasets in a given repository. Within the registry, a federated data catalog system provides a single point of access across all data regardless of the location data is stored. The use of Archival Resource Keys (ARKs) as persistent identifiers (PIDs) means that specific data can be queried and accessed from a request originating outside the system. Using Nice Opaque Identifier (NOID) software tool to implement ARKs provides a way to mint unique ARK strings, bind location URLs to those strings, and serve as a name resolver to redirect requests to the proper location of each dataset. We explain the method details in this section.

2.1 Federated Data Types Registry

Automated processing of large volumes of data requires implicit details about the data types. Data federation combines all the autonomous data stored to

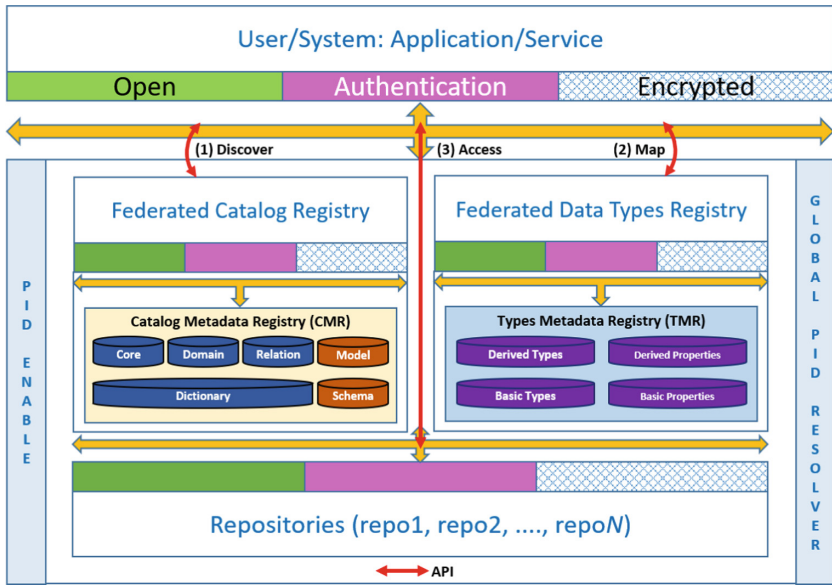


Fig. 1. Block diagram of a federated metadata registry (BDGMMWG Draft Fig. 1 [4]).

form one extensive database. Data federation stores data into a heterogeneous set. The data is accessible to users as one integrated database using on-demand data integration [10]. A data user using data federation should be able to access different types of database servers and files with various formats from all data sources and be accessible through multiple APIs.

Types Metadata Registry (TMR). A metadata registry is a database used to store, organize, manage, and share metadata [2]. An issue often encountered in building an extensive metadata registry is that the participants may have used diverse schemas and description methods to create their metadata records [3]. Building one integrated metadata registry enables one search from the user's end to retrieve the information needed rather than searching through all the different existing individual databases. Therefore, interoperability will be a critical component in such development. Most of the time, different types of resources are already classified under a variety of specialized schemas [3].

Data Types are characterizations of data at any level of granularity. Data types identified, defined and registered data. When encountering unknown data, the registry can explicate those types and output the type definitions, relationships and properties. *Basic Types* are used for interpreting and later processing data. A Data Type Registry provides a detailed and structured description of inputted data. Some of the basic types are listed below in Table 1 (left column). *Derived Types* are more specific cases/instances of the basic types. For example, the derived types are listed in Table 1 (right column).

Table 1. Derived Types

Basic Types	Corresponding Derived Types
text	character, varchar
number	integer, long, real, float, double, percentage, scientific
currency	USD, RMB, \$, Euro, Yen
boolean	check box, yes/no, true/false, on/off
date/time	timestamp, short date, medium date, long date, time am/pm, medium time, time 24 h
blob	rich text, attachment, memo, attachment
calculated	lambda unction, imaginary number
pointer	hyperlink, lookup

Basic Properties describe the nature of the data. These descriptions are essential to understanding and performing any statistical analysis on the data. Different Exploratory Data Analysis (EDA techniques) can be used to identify the properties of data so that the appropriate statistical methods can be applied to the data. Here is a list of different basic properties of a set of data as shown in Table 2 *Derived Properties* are properties that do not exist on the Entity Type associated with the Entity Set; rather, it exists on a type that is derived from the base type of the entity set.

Table 2. Basic Properties

Basic Properties
Centre of data
Skewness of data
Spread among the data members
Presence of outliers
Correlation among the data
Type of probability distribution that the data follows

2.2 Federated Catalog Registry

A data catalog system provides a single place where all available data can be catalogued, enriched, searched, tracked and prioritized. Federation provides a single view across all data of interest to a user, regardless of where the data is stored or sourced. When there are changes to the external data, the system can quickly crawl external data sources, track changes, make automatic enhancements and push notifications when changes occur. A federated registry is used when user and group information is spread across multiple registries. If the information is

stored in various systems, a federated registry can provide a unified view without changing the platform providing a single point of access to reliable data.

Catalog Metadata Registry (CMR). A metadata Catalog System provides a single place where all data can be catalogued, enriched, searched, tracked and prioritized. Data catalog is a collection of metadata, combined with data management and search tools. The combination serves as an inventory for data and provides tools for data analysis. The analyst can search and find data quickly, see all of the available datasets, evaluate and make informed choices for which data to use, and perform data preparation and analysis efficiently.

Core is the main classifications and groups among the Catalog Metadata Registry. *Domain* has three types: environmental domain, object class domain and object format domain. Environmental domain is the discipline and the community that the scheme services. Object class domain is the assembly and grouping of similar objects by type and offer multiple ways to define type. Object format domain is the object's composition and what it is made of [6]. *Relation* defines a related resource. The recommended practice is to identify the related resource by means of a URI. If this is not possible or feasible, a string conforming to a formal identification system may be provided [1]. *Model* describes the physical raw data files such as: binary, images, or text containing numeric values [11]. *Dictionary* is a set of well defined terminology. The goal for a dictionary is to eliminate ambiguity, control the use of synonyms, establish formal relationship among terms and validate terms [7]. *Schema* is also called Formats or Element Sets. It is a set of semantic properties used to describe resources; examples include: MARC 21, Dublin Core, MODS, and ONIX [7].

2.3 ARKs as Persistent Identifiers (PIDs)

A Persistent identifier (PID) is a long-lasting reference/resolve to a digital resource. There are many online organizations that provide PID resolving services; for example, DOIs, ORCIDs, and ARKs. Digital object identifiers (DOI, <https://www.doi.org>) are paid-for persistent identifiers used for scholarly communication, such as in journal articles, books and datasets. DOIs as a paid for service. Open Researcher and Contributor Identifiers (ORCID, <https://orcid.org>) is financially sustained by membership fees from organizational members; however, participants in research, scholarship or innovation can register an ORCID ID for free. Archival Resource Keys (ARKs, <https://arks.org>) are persistent identifiers (PIDs) that are open, decentralized and can be applied to any type of object (digital, physical or abstract). Like any modern PID system, ARKs rely heavily on URL redirection. To use ARKs, the following items are needed: a Name Assigning Authority Number (NAAN), a minter, a resolver, an assignment plan, and an access persistence policy.

The NAAN indicates the organization that creates the persistent identifier. To eliminate duplicates, the minter only generates unique string. Once a string

is assigned to a resource and its location, the resolver can redirect the persistent ARK identifier to the current access URL where the resource resides. Pre-planning of ARK features such as suffix pass-through (which allows one ARK to be a single point of management for millions of other ARKs) or ARK shoulders (a way of subdividing and managing your own namespace) should be considered during the implementation. If the resource is kept with persistent access, it needs to be securely preserved to prevent accidental loss and deletion [9].

The Nice Opaque Identifier (NOID) tool lets one establish any number of minters. Each NOID minter comes with a template specifying the length and combinations of numbers and letters in the strings that it will generate. These strings become ARKs when they are assigned to the objects they identify. In order to prevent duplication, a database will check that no identifier is minted more than once. The published URL is kept persistent while NOID can also be used to run as a name resolver to redirect to the new location of where the object resides. You can publish your ARKs with your own hostname (decentralized ARKs) or with the N2T.net hostname (centralized mode, in case your own hostname is subject to some instability).

3 Designing PIDs for Time-Series Data

Time-series data is a collection of observations made chronologically. For example, where sensor measurements are recorded over a period of time. Given the length of time and the frequency of reading, these datasets can become large in size, and have high dimensionality if reading more than one value, observation, or measurement [5]. Each value from each sensor can be marked with a timestamp — the date and time of reading — a unique identifier. As the timestamp is unique we can use it to reference a specific reading. Row number cannot be used as a unique identifier because it can change based on data manipulation and information processing functions such as row sorting.

For our demonstration we use the AMPds: Almanac of Minutely Power dataset [8] which contains electricity, water, and natural gas metering data. Sensors measure different consumption characteristics. In each CSV file (one per sensor) the timestamp (per minute) is the a unique identifier or primary key.

An example of a PID might be:

Ex.1: <https://n2t.net/ark:/57460/AMPds/DWE/V@13332~13400>

where [https](#) is a secure HTTP RESTful request, [n2t.net](#) is the global resolver, [ark](#) means the URL is an ARK, [57460](#) is the NAAN for our lab, [AMPds](#) is the name of the dataset, [DWE](#) is the meter or sensor, [V](#) is the measurement (column), and [@13332~13400](#) is the timestamp range of the readings (rows). To select all the readings for a given sensor (e.g., DWE or dishwasher meter) for a give measurement (V or voltage), the following URL would be used:

Ex.2: https://n2t.net/ark:/57460/AMPds/DWE/V@*

or multiple timestamp (i.e., ID) ranges we would use @13332~13400+24300~25500. To return multiple values (e.g., V for voltage and I for current) we use:

Ex.3: https://n2t.net/ark:/57460/AMPds/DWE/V+I@*

And, for multiple sensors (e.g., HPE, DWE, and WOE; heatpump, dishwasher, and wall oven) we would use:

Ex.4: https://n2t.net/ark:/57460/AMPds/HPE+DWE+WOE/V+I@*

Lastly, using cross-fold validation is important for training and testing learning algorithms. Here is an example when we want to exclude a range of timestamps:

Ex.5: <https://n2t.net/ark:/57460/AMPds/DWE/V@@24300~25500>

In this case, we are requesting all data except for the data in the inclusive timestamp range of @@24300~25500. The double-at @@ is used as *exclusion* rather than the minus sign as this avoids potential URL corruption with text formatting software where hyphens as used for line breaks.

4 Conclusions

We show how PIDs can be used for persistent access to specific data from a dataset. ARKs provides a flexible, inexpensive way to make data more accessible and provide an easy way to describe the data for easy reproducibility of research and scientific results. Next steps include releasing a prototype that can be used as part of the IEEE Standards Association P2957 Data Governance and Metadata Management Working Group (BDGMMWG) standard. Further, we plan to demonstrate using PIDs for other types of non-time-series data.

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